System of Linear Equations

Ex- The sum of any two of three real numbers are 24, 28 and 30. Find these three real numbers.

Hints of Solⁿ :-

$$\left.\begin{aligned} x+y &= 24 \\ y+z &= 28 \\ x+z &= 30 \end{aligned}\right\}$$

Now find the value of x, y, z.

The need to solve system of linear equations frequently aries in mathematics, statistics, physics, astronomy, engineering, computer science, economics, etc.

Solving systems of linear equations is not conceptually difficult. For solving small systems any ad hoc methods works well. For large systems we need more systemtic method. The method we use today is similar to a method used in 2,000 years ago in China.

Main Idea :- Ex-

$$\left|\begin{aligned} x+2y+3z &= 14 \\ x+3y+2z &= 13 \\ 3x+2y+z &= 10 \end{aligned}\right|$$

To solve this system we want to tranform the system into the form

$$\left|\begin{aligned} x \quad\quad &= \cdots \\ y \quad &= \cdots \\ z &= \cdots \end{aligned}\right|$$

That is, we need to eleminate the tepm that are off diagonal, (circled in the above equation) and make to coefficients of the variables along the diagonal equal to 1.

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NB: The system is also easy to solve if we can transform the system as

$$\left| \begin{array}{r} x + 2y + 3z = 14 \\ y - z = -1 \\ 12z = 36 \end{array} \right|$$

Then the solution can be easily obtained by eleminating the variables iteratively as follows.

$$12z = 36 \Rightarrow z = 3$$

$$y - z = -1 \Rightarrow y - 3 = -1 \Rightarrow y = 2$$

$$\cancel{x + 2y + 3z = 14 \Rightarrow x + 6 +}$$

$$x + 2y + 3z = 14 \Rightarrow x + 2.2 + 3.3 = 14 \Rightarrow x = 1$$

Possible outcomes

(a) No solution
The system is inconsistent.

Has solution(s)
System is consistent

(b) Infinitely many solutions.

(c) Exactly one solution.

Geometric Interpretation

Ex-1 Two straight lines in 2-D

(a) No solⁿ : || lines

$$2x + 3y = 5$$
$$4x + 6y = 2$$

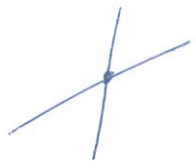
A solution i.e., a common point
There is no point lies on ~~the~~ both the st. lines

(b) Infinitely many ~~solution~~ solⁿ :

$$2x + 3y = 5$$
$$4x + 6y = 10$$

Since both the lines are the same, there are infinitely many points lie on both the lines.

(c) Exactly one solⁿ :



Two intersecting st. lines.
There is exactly one common point.

Ex-2 Three Planes in 3-D

Note: (a) Two planes intersect at a st. line.
(b) One plane and one line intersect at a point.

(a) No Sol^n: (i) Three parallel planes

(ii) Two parallel planes and ~~one~~ another plane intersects ~~with~~ the other two planes

(iii) Each pair of planes intersect at three different lines.

That is there is no point which lies on these three planes

(b) Infinitely many Sol^n: These three planes intersect at a common st. line.

All the points of that st. line lie on these three planes. Since there are infinitly many points on a st. line, there are infinitely many sol^n of this system.

(c) Two planes intersect at a st. line and the third plane intersects with that st. line at a point.

That (~~sol~~ (single) point lies on these three planes simultaneously.

Matrix

Ex: 1

$$x + 2y + 3z = 16$$
$$4x + 5y + 6z = 35$$

$$\rightarrow \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}_{2\times 3}$$

coefficient matrix

Ex-2

$$2x + 8y + 4z = 2$$
$$2x + 5y + z = 5$$
$$4x + 10y - z = 1$$

Coefficient Matrix $\rightarrow \begin{bmatrix} 2 & 8 & 4 \\ 2 & 5 & 1 \\ 4 & 10 & -1 \end{bmatrix}_{3\times 3}$

~~Genery~~ General representation

$$\begin{bmatrix} a_{11} & a_{12} & a_{13} & \cdots & a_{1n} \\ a_{21} & a_{22} & a_{23} & \cdots & a_{2n} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & a_{m3} & \cdots & a_{mn} \end{bmatrix}_{m\times n} =: A$$

(*) If $m = n \Rightarrow A$ is a square matrix.

(*) If $a_{ij} = 0 \ \forall\, i > j \Rightarrow A$ is an upper triangular matrix

Ex - (a) $\begin{bmatrix} 1 & 2 & 3 \\ 0 & 4 & 1 \\ 0 & 0 & 6 \end{bmatrix}$ (b) $\begin{bmatrix} 2 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 0 \end{bmatrix}$

(*) If $a_{ij} = 0 \ \forall\, i < j \Rightarrow A$ is an lower triangular matrix.

Ex - $\begin{bmatrix} 5 & 0 & 0 \\ 4 & 0 & 0 \\ 3 & 2 & 1 \end{bmatrix}$

Ex -

(X) If $m = n$ and $a_{ij} = 0 \ \forall \ i \neq j \Rightarrow A$ is diagonal

Ex - $\begin{bmatrix} 2 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 0 \end{bmatrix}$

Ex - (3)

$$2x + 5y + z = 5$$
$$2x + 8y + 4z = 2$$
$$\cancel{2x + 5y + z = 5}$$
$$4x + 10y - z = 1$$

Augmented Matrix $\longrightarrow$ $\left[\begin{array}{ccc|c} 2 & 5 & 1 & 5 \\ 2 & 8 & 4 & 2 \\ 4 & 10 & -1 & 1 \end{array}\right]$

Method: Rules of operations ~~for~~ to obtain the solution!

Elementary Row Operations

1. (Row swap): Exchange any two rows.
2. (Scalar multiplication): Multiply any row by a constant.
3. (Row sum): Add a multiple of a row to another row.

Ex-

$$\left[\begin{array}{ccc|c} 2 & 5 & 1 & 5 \\ 2 & 8 & 4 & 2 \\ 4 & 10 & -1 & 1 \end{array}\right]$$

$$\underset{\sim}{R_2 \leftrightarrow R_1} \quad \left[\begin{array}{ccc|c} 2 & 8 & 4 & 2 \\ 2 & 5 & 1 & 5 \\ 4 & 10 & -1 & 1 \end{array}\right]$$

$$\underset{\sim}{\tfrac{1}{2}R_1} \quad \left[\begin{array}{ccc|c} 1 & 4 & 2 & 1 \\ 2 & 5 & 1 & 5 \\ 4 & 10 & -1 & 1 \end{array}\right]$$

$$\underset{\sim}{R_3 = R_3 - 4R_1} \quad \left[\begin{array}{ccc|c} 1 & 4 & 2 & 1 \\ 2 & 5 & 1 & 5 \\ 0 & -6 & -9 & -3 \end{array}\right]$$

Method 1

* **Leading entry in a row**: If a row has nonzero entries, then the first non-zero entry is called the leading in that row.

Goal: Transform the augmented matrix into Row-Echelon Form (REF) (by using elementary row operations).

· **Row-Echelon Form (REF)**: A matrix is in REF if has the following properties.

(a) All non zero rows are above any rows of all zeros.

(b) If a row contains a leading entry, then each row above it contains a leading entry further to the left.

(c) All ~~entries~~ entries in a column below a leading entry are zeros.

Transforming a matrix into REF (by using elementary row operations) is called Gauss elimination method.

EX-

$$\begin{bmatrix} 0 & 2 & 1 & 0 & 5 & 9 \\ 0 & 0 & 0 & 5 & 2 & 7 \\ 0 & 0 & 0 & 0 & 3 & 6 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}, \quad \begin{bmatrix} 2 & 1 & 0 & 0 \\ 0 & 0 & 3 & 4 \\ 0 & 0 & 0 & 2 \\ 0 & 0 & 0 & 0 \end{bmatrix}, \quad \begin{bmatrix} 2 & 6 & 5 \\ 0 & 1 & 3 \\ 0 & 0 & 4 \end{bmatrix}$$

Note: All are upper triangular matrix

History of Gauss elimination method

It is named on the German mathematician Carl Friedrich Gauss (1777 – 1855).

This method is based on a method used by chinese 2000 years ago.

When Gauss were 17 years old, he developed a method for working with the inconsistent linear systems, called the method of least squares.

At his age of 24 years, he turned his attention to a particular problem in astronomy. On January 1, 1801, the Sicilian astronomer Giuseppe Piazz (1746 – 1826) discovered a ~~planet~~ (dwarf) planet, which he named Ceres, in honor of the patron goddess of Sicily. Today Ceres is called a dwarf planet, because it is only about 1,000 kilometers in diameter. Piazzi was able to observe Ceres for 40 nights, but then lost track of it when it passed behind the sun. Piazzi had only tracked Ceres through about 3 degrees of sky. Gauss, however, succeeded in calculating the ~~of~~ orbit of Ceres, even though the task seemed hopeless on the basis of a few observations. His computations were so accurate that the German astronomer W. Olbers (1758 – 1840)

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located the planet on December 31, 1801. In the course of his computations, Gauss had to solve systems of 17 linear equations. Eight years later, in 1809, Gauss revealed his method of orbit computation in his book, Theoria Motus Corporum Coelestium.

Ex of Gauss elimination method

Ex-1

$$3x + 2y + z = 10$$
$$2x + 4y + 6z = 28$$
$$x + 3y + 2z = 13$$

First we write the system as an augmented matrix:

$$\left[\begin{array}{ccc|c} 3 & 2 & 1 & 10 \\ 2 & 4 & 6 & 28 \\ 1 & 3 & 2 & 13 \end{array}\right] \underset{\sim}{R_1 \leftrightarrow R_2} \left[\begin{array}{ccc|c} 2 & 4 & 6 & 28 \\ 3 & 2 & 1 & 10 \\ 1 & 3 & 2 & 13 \end{array}\right]$$

$$\underset{\sim}{\tfrac{1}{2}R_1} \left[\begin{array}{ccc|c} 1 & 2 & 3 & 14 \\ 3 & 2 & 1 & 10 \\ 1 & 3 & 2 & 13 \end{array}\right] \underset{\sim}{R_2 = R_2 - 3R_1} \left[\begin{array}{ccc|c} 1 & 2 & 3 & 14 \\ 0 & -4 & -8 & -32 \\ 1 & 3 & 2 & 13 \end{array}\right]$$

$$\underset{\sim}{R_3 = R_3 - R_1} \left[\begin{array}{ccc|c} 1 & 2 & 3 & 14 \\ 0 & -4 & -8 & -32 \\ 0 & 1 & -1 & -1 \end{array}\right] \underset{\sim}{R_3 = R_3 + \tfrac{1}{4}R_2} \left[\begin{array}{ccc|c} 1 & 2 & 3 & 14 \\ 0 & -4 & -8 & -32 \\ 0 & 0 & -3 & -9 \end{array}\right]$$

Now we are in REF. The associated equations are

$$x + 2y + 3z = 14 \Rightarrow x + 2(2) + 3(3) = 14 \Rightarrow x = 1$$
$$-4y - 8z = -32 \Rightarrow -4y - 8(3) = -32 \Rightarrow y = 2$$
$$-3z = -9 \Rightarrow z = 3$$

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